Chronotype's Effect on Mental and Physical Health Across Countries

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Abstract

Chronotype refers to an individual's inherent preference for the timing of their sleep and wakefulness, and previous studies have linked differences in chronotype to various health outcomes. This study investigated the relationship between chronotype and mental, physical, and depression levels at both the within-country and between-country levels. We recruited 5,128 participants from 20 countries. They completed the Munich Chronotype Questionnaire and Patient-Reported Outcomes Measurement Information System (PROMIS) scales to assess their chronotype and health outcomes, including mental health, physical health, and depression. Multilevel modeling was conducted to determine the effects of chronotype on each outcome, within and between countries. Results showed that at the within-country level, later chronotypes were associated with lower mental health and physical health scores but higher depression scores, as measured by PROMIS scales. No significant association was observed at the between-country level. Our study found a significant association between chronotype and health outcomes at the within—but not the between—country level, suggesting that people's chronotype may only be associated with health outcomes relative to their own country's average chronotype. Our study builds on prior findings by highlighting the importance of considering within-country variations in chronotype when examining health outcomes.

Keywords: sleep; chronotype; morningness-eveningness; depression, physical health; cross-countries study.

Circadian rhythm is a universal endogenous process that determines a person's sleep-wake cycle (Czeisler & Gooley, 2007). A chronotype is the behavioral manifestation of the underlying circadian rhythm, varying across individuals (Adan et al., 2012). Specifically, a person's chronotype is their tendency to sleep at a particular time during a 24-hour period (Natale & Cicogna, 2002). There are two general types of chronotypes: morning chronotype and evening chronotype (Adan et al., 2012). People with the morning chronotype usually rise early and are more active in the morning, while those with the evening type typically wake up later and are more active in the afternoon or evening (Adan et al., 2012). However, most people fall somewhere between these two types and have some flexibility (Natale & Cicogna, 2002). Consequently, it is more acceptable in the literature to view chronotype as a continuum rather than two extremes (Natale & Cicogna, 2002).

Past research has highlighted associations between chronotype and health outcomes, revealing that evening chronotypes generally experience worse health outcomes than morning chronotypes (Fabbian et al., 2016; Merikanto et al., 2013; Randler et al., 2017). These studies also explore cultural differences in chronotype distribution across countries, emphasizing the importance of considering geographical location, latitude, and sunrise time as possible factors influencing these differences (Adan et al., 2012; Randler, 2008a; Duffy et al., 2011; Borisenkov, 2011). However, the effects of chronotype on health have shown inconsistent results across different countries (Figueiro et al., 2014; Haraszti et al., 2014; Levandovski et al., 2011; Mongrain et al., 2006). This inconsistency raises the possibility that the impact of chronotype on health may be country-specific or more significant within countries, rather than a universal phenomenon. The present research seeks to elucidate this relationship by examining the effects of chronotype on health within different countries, aiming to clarify whether these effects are indeed

country-specific or universally applicable.

Chronotypes Across Countries

Extensive studies have investigated the differences in chronotypes of people in different countries throughout the world (Adan et al., 2012; Randler, 2008a; Duffy et al., 2011; Borisenkov, 2011). For instance, Adan et al. (2012) examined the chronotypes of individuals from 12 countries, including North America, Europe, Asia, and Africa, and found significant differences in chronotype distribution across these countries. Similarly, Randler (2008a) compared adolescents from Germany, Turkey, and other countries, identifying differences in chronotype preferences between these populations. In a study conducted by Horzum et al. (2015), researchers developed the environmental theory suggesting that people living near the equator have a higher morning orientation than those living further from the equator. This theory was supported by comparisons of chronotypes between Turkish and German adolescents. Randler (2008b) also supported this theory by finding that adolescents in the subtropics were the latest chronotypes, while adolescents in the tropics area were the earliest chronotypes. Another factor found to affect chronotype differences across countries is sunrise time. Randler (2008c) discovered that adolescents in the east of Turkey were more morning-oriented than adolescents in the west, suggesting that people residing in places where the sun rises earlier will be more morning-oriented than people residing in places where the sun rises later. This finding was further confirmed by Roenneberg et al. (2007), which states that sunlight acts as the entrainment for the biological clock. Moreover, Duffy et al. (2011) found that, in a sample of 537 individuals from 5 different countries, there were substantial differences in the distribution of chronotypes, with individuals from the USA and the Netherlands showing the latest chronotypes compared to those from Japan, Germany, and Brazil. A study by Borisenkov (2011) also supported the

existence of differences in chronotype distribution, reporting that Russians were more morning-oriented than the French. These studies have provided evidence of differences in chronotypes across countries, highlighting the importance of considering geographical location, latitude, and sunrise time as possible factors influencing these differences (Adan et al., 2012; Randler, 2008a; Duffy et al., 2011; Borisenkov, 2011).

Chronotype, Health Outcomes, and Cultural Differences

In addition to the fact that the average chronotype differs across countries, the relationship between chronotype and health outcomes also differs across countries, with studies in some countries showing significant association between late chronotype and poor health outcomes (e.g., Partonen, 2015; Vetter et al., 2016) while studies in other countries finding no significant results (e.g., Figueiro et al. 2014). For example, with North American samples, evening chronotypes have been found to have significantly higher odds of developing physical illnesses such as cardiovascular disease (Vetter et al., 2016), cancer (Papantoniou et al., 2015), and metabolic syndrome (Ko et al., 2019). In terms of mental health, with a sample from Finland, Partonen (2015) demonstrated that people with an evening chronotype are more likely to suffer from various mental disorders, including anxiety disorders, substance use disorders, personality disorders, and depression; and with a sample from Australia, Au and Reece (2017) suggested that an evening chronotype might serve as a risk factor associated with depressive symptomatology by leading individuals to engage in harmful behaviors such as insomnia, which could contribute to depression. Some recent research has also investigated the potential mechanisms underlying the association between late chronotype and poor health outcomes (Leproult et al., 2014; McGowan et al., 2016). For instance, Leproult et al. (2014) found that evening types exhibited reduced insulin sensitivity, suggesting a possible biological basis for the increased risk of

diabetes in this population. Additionally, McGowan et al. (2016) demonstrated that evening types had higher levels of cortisol, a stress hormone, which could explain the increased risk of mood disorders and other negative health outcomes in this group. Sleep quality, as influenced by chronotype, has also been linked to health outcomes (Kanerva et al., 2018; Fabbian et al., 2016). For example, Kanerva et al. (2018) observed that evening types experienced poorer sleep quality compared to morning types, which was associated with an increased risk of obesity and type 2 diabetes. Similarly, Fabbian et al. (2016) found that evening types had a higher prevalence of sleep disorders such as insomnia and sleep apnea, which may contribute to the observed differences in health outcomes between chronotypes.

Although the aforementioned studies all showed that there is a strong association between late chronotype and poor health outcomes, there are also many studies that found weaker or no relationship between late chronotype and poor health outcomes (Figueiro et al., 2014; Haraszti et al., 2014; Levandovski et al., 2011; Mongrain et al., 2006). For example, in a study conducted in Portugal, Figueiro et al. (2014) found no significant association between chronotype and mental health. The researchers investigated the association between chronotype and factors such as mood, sleep quality, and daily functioning, but the results did not support the hypothesis that evening chronotypes were more likely to experience poorer psychological well-being than morning chronotypes (Levandovski et al., 2011). Similarly, a study by Haraszti et al. (2014) on Hungarian university students also reported non-significant associations between chronotype and measures of psychological distress, suggesting that the relationship between chronotype and health outcomes might not be consistent across all populations. A study by Levandovski et al. (2011) in Brazil reported weaker associations between chronotype and health outcomes. Their study included measures of mood, sleep quality, and social adjustment, and although some

significant relationships were identified, the overall strength of the associations between chronotype and health outcomes was less pronounced compared to the findings in other countries. Furthermore, a study by Mongrain et al. (2006) conducted in Canada found no significant relationship between chronotype and measures of mental health, such as depression, anxiety, and stress. This study adds to the growing evidence of inconsistent findings across different countries regarding the association between chronotype and health outcomes. These inconsistent findings across different countries highlight the need for further research to determine whether the impact of chronotype on health outcomes is country-specific or universally applicable.

The Present Research

The present research aims to elucidate this relationship between chronotype and health outcomes by examining the effects of chronotype on health outcomes within and between different countries, clarifying whether these effects are country-specific or universally applicable. Our study was conducted on a large scale, examining the relationship between chronotype and health outcomes with participants in 20 countries across six continents. By employing multilevel modeling analyses, we aim to investigate the within-country and between-country effects of chronotype on health outcomes, which would provide a more comprehensive understanding of the relationship between chronotype and health outcomes. Based on the inconsistent findings in the literature (e.g., Fabbian et al., 2016; Merikanto et al., 2013; Randler et al., 2017), some showing significant relationships between chronotype and health outcomes while others not, we hypothesize that the effect is only country-specific; in other words, the relationship between chronotype and health outcomes is significant within countries but not between countries.

Method

Participants

We recruited 5128 participants, originating from 20 countries across six continents, with approximately 250 participants from each country. All participants are of domestic cultural heritage. Table 1 shows the number of participants in each of the 20 countries. We chose 20 countries because Monte Carlo simulation of multilevel estimates and confidence interval coverage of estimated macro effects indicated that the inclusion of 15 to 20 countries is sufficient to limit bias for between-country effects (Stegmueller, 2013). In addition, for the within-country effect, Monte Carlo simulation indicated that a sample size of more than 250 participants allows for stable estimates in typical scenarios (Schonbrodt & Perugini, 2013). After excluding participants who answered having a nightshift job or declined to participate (N = 187), the final sample size was 4941. The mean age of the participants is 39.41 (SD = 9.76). Out of the 4941 participants, 40.54% identified themselves as men, 59.24% as women, and 0.14% as non-binary.

Materials

Participants completed a survey on the Qualtrics platform. The survey responses were all collected during a window that extended ten days before and after the autumnal equinox when the length of the day was similar across the globe. In this survey, participants answered various questions about their chronotype and health outcomes. All materials were translated from English to other languages by translators with native backgrounds, and then the translations were further examined by one of the authors.

MCTQ Chronotype

Participants answered a subset of questions from the Munich Chronotype Questionnaire (MCTQ) to assess their chronotypes (Roenneberg et al., 2003). It should be noted that not all

questions from the full MCTQ were included in the survey, potentially impacting the accuracy and comprehensiveness of the measurement. We only included objective measures of sleep and wake-up times on workdays and free days, excluding the subjective ideal sleep and wake-up times. The MCTQ is a self-report measure that examines individuals' sleep-wake patterns on both workdays and free days, asking about their actual sleep and wake-up times. Based on this information, a participant's chronotype is calculated as the midpoint of sleep on non-workdays, adjusted for sleep debt accumulated during workdays (referred to as MSFsc). In this study, the MSFsc is determined by calculating the number of minutes from the middle point of the sleep from the previous day's noon (i.e., 12 pm). Lower MSFsc values indicate earlier chronotypes, while higher values represent later chronotypes. The MCTQ has demonstrated good validity and reliability for assessing chronotype in various populations (Zavada et al., 2005).

PROMIS Global Health

The PROMIS Global Health scale was used to measure an individual's physical and mental health (Cella et al., 2010). This scale is an 8-item form with four items measuring physical health (e.g., "In general, would you say your physical health is excellent, very good, good, fair, or poor?") and four items measuring mental health (e.g., "In general, how often have you been bothered by emotional problems such as feeling anxious, depressed, or irritable?"). Each item on the measure is rated on a 5-point scale, with higher scores indicating worse health levels. To compute the physical and mental health scores, the raw scores for each subscale (physical and mental health) were summed separately and then transformed into *t*-scores using the appropriate scoring manual provided by PROMIS Health Organization (PHO) and PROMIS Cooperative Group. According to the manual, the *t*-scores are standardized scores in comparison with the US population with a mean of 50 and a standard deviation of 10, with higher *t*-scores

indicating better health. Using this scale, we can compute one global physical health *t*-score and one global mental health *t*-score for each participant.

PROMIS Depression

The PROMIS Depression scale was used to measure depression (Pilkonis et al., 2011). This scale is an 8-item form that evaluates various aspects of depression, such as negative mood, decrease in positive affect, and cognitive and somatic symptoms (e.g., "In the past 7 days, I felt hopeless."). Each item asks participants to rate the severity of their depression in the past seven days. These items are rated on a 5-point scale (1 = never; 2 = rarely; 3 = sometimes; 4 = often; and 5 = always), with higher scores indicating greater severity of depression. To compute the depression score, the raw scores are summed and then transformed into standardized *z*-scores using the scoring guidelines provided by the PROMIS PHO and PROMIS Cooperative Group. The *z*-scores are normed based on the US population, with a mean of 0 and a standard deviation of 1. Higher *z*-scores indicate greater severity of depression. Using this scale, we can compute a depression *z*-score for each participant.

Procedure

Participants were recruited via a Qualtrics survey panel in 20 countries. Participants were sent the invitation or prompted on the respective survey platform to proceed with a given study. When participants were invited to take a survey, they were informed that they would be compensated, and the rate of remuneration is determined per country by Qualtrics. In this survey, participants provided their demographic information (e.g., sex, gender) and general health (e.g., weight). Participants also provided information regarding their actual sleep and wake-up times. Next, they answered questions about measuring their actual chronotype. Finally, they answered questions regarding their physical and mental health.

Statistical Analyses

We conducted the multilevel modelling analyses to examine the within and between countries effect of chronotype on the physical and mental health outcomes. We used the lme4 package (De Boeck et al., 2011) in the *R* software to do the multilevel modelling analyses. Within-country chronotypes (i.e., centered within countries) and country average chronotypes were used as predictors in the models. Separate analyses were performed for each of the three outcome variables: 1) global mental health measured by PROMIS *t*-score, 2) global physical health measured by PROMIS *t*-score, and 3) depression measured by PROMIS *z*-score. Outliers, defined as those who answered less than one of the outcome variables, were excluded from the analyses. Demographics variables, such as gender, age, education and income, were statistically controlled for in the multilevel modelling analyses.

Results

Descriptive

Table 2 shows the descriptive statistics on variables of interest (e.g., chronotype, PROMIS depression score) both for the whole sample and individual countries. Table 2 displays the means and standard deviations of chronotype, PROMIS depression *t*-scores, PROMIS mental health *t*-scores, and PROMIS physical health *t*-scores for each country. On average, participants from Argentina had the latest evening chronotype with a mean MSFsc of 1014.08 mins (*SD* = 86.19 mins), whereas participants from Colombia had the earliest chronotype with a mean MSFsc of 897.98 mins (*SD* = 79.88 mins).

Multilevel Modelling Analyses

Tables 3, 4 and 5 show the results from multilevel modelling analyses using global physical health, global mental health, and depression as the outcome variable, respectively.

Figure 1 shows the within and between countries' effects of chronotype on global physical health, global mental physical, and depression. Overal, the results show that within countries, people with chronotypes that were later than their countries' averages were more likely to have lower global physical *t*-score (b = -0.54, 95%CI = [-0.79, -0.33], $p < 0.001$) , lower global mental *t*-score (b = -0.75, 95%CI=[-1.08, -0.42], $p < 0.001$), and higher depression *z*-score (b = 0.07, 95%CI=[0.04, 0.11], p = 0.001; see Tables 3-5 and Figure 1). In contrast, between countries, there were no significant relationships between chronotype and each of the outcome variables (see Tables 3-5 and Figure 1).

Discussion

The present study aimed to investigate the association between chronotype and health outcomes both within and between countries. Our results supported our hypothesis that people with later chronotypes than their country's average were associated with lower scores of physical and mental health outcomes as measured by PROMIS scales. Specifically, people with later chronotypes had lower scores on the PROMIS global physical and mental scales, but higher scores on the PROMIS depression scale, compared with those with earlier chronotypes relative to their country's average. However, no significant relationship was found between chronotype and health outcomes between countries on any of the three measures. These results highlight the importance of considering the relative chronotype when examining the link between chronotype and health outcomes, indicating that people's chronotypes are most predictive of their health outcomes when compared to their own country's average.

Our study adds to the existing literature on chronotype's effects on people's health by examining the relationship in a country-specific context. Building upon the previous inconsistent results, our findings suggest that the impact of chronotype on health outcomes is determined by

its relative position within the cultural context. This highlights the importance of considering cultural differences when examining the relationship between chronotype and health.

One possible reason underlying the difference between within- and between-country effects is that the relative chronotype, or how an individual's chronotype compares to the average in their own country, plays a crucial role in determining health outcomes (Levandovski et al., 2011; Fischer et al., 2017). This is because individuals with later chronotypes relative to their country's average may experience a greater discrepancy between their natural sleep preferences and the social expectations of their country, leading to poorer health outcomes (Wittmann et al., 2006; Randler, 2018). Within a country, social norms and expectations are more likely to be consistent, so the relative chronotype becomes an important factor in determining the impact on health outcomes (Borisenkov, 2011; Roenneberg et al., 2019). Individuals who deviate more significantly from their country's average chronotype may experience a greater misalignment between their natural sleep-wake patterns and the societal norms, which could lead to higher levels of stress, sleep disturbances, and ultimately poorer health outcomes (Knutson & von Schantz, 2018; Vetter et al., 2018). In contrast, the between-country analysis may not accurately capture the nuances of relative chronotype, as it does not account for the wide variation in social norms, average chronotypes, and the degree to which individuals deviate from their country's average (Levandovski et al., 2011; Fischer et al., 2017). The between-country effect may be insignificant due to the substantial differences in social norms and average chronotypes across countries, weakening the association between chronotype and health outcomes across different cultural contexts (Roenneberg et al., 2019; Borisenkov, 2011). To better understand the relationship between chronotype and health outcomes within and between countries, future research should investigate the role of relative chronotype while accounting for cultural

differences and variations in social norms related to sleep patterns (Levandovski et al., 2011; Randler, 2018). In addition, researchers could explore how discrepancies between individuals' natural sleep preferences and their country's social expectations may contribute to the differences in health outcomes observed in the within- and between-country analyses (Wittmann et al., 2006; Knutson & von Schantz, 2018). This would help develop culturally sensitive interventions to promote healthy sleep habits and improve health outcomes across diverse populations (Vetter et al., 2018; Roenneberg et al., 2019).

Limitations

Several limitations must be considered when interpreting the findings of this study. First, although the Munich Chronotype Questionnaire (MCTQ) is a validated measure of chronotype, some questions were not included in the survey, potentially impacting the accuracy and comprehensiveness of the measurement (Roenneberg et al., 2003). Future studies may benefit from the use of the full MCTQ scale or other validated measures, such as the Morningness-Eveningness Questionnaire (MEQ) (Horne & Östberg, 1976), to improve the reliability of the results. Second, the sample size for each country was relatively small, with only 250 participants per country, which may limit the generalizability of the results and increase the risk of sampling bias (Schonbrodt & Perugini, 2013). It is important to exercise caution when generalizing the findings to other populations, as the participants in the study may not be representative of the entire population of their respective countries. Third, as the study was not preregistered, the results should be interpreted with caution, and further research is needed to confirm the findings (Wagenmakers et al., 2012). Finally, the *t*-scores used in the PROMIS global scales for measuring physical and mental health were computed in comparison with the US population mean and standard deviation (Cella et al., 2010). As a result, the *t*-scores may not

accurately reflect the health outcomes of participants from other countries or cultures.

Conclusion

In summary, our study conducted across multiple countries revealed a significant correlation between chronotype and health outcomes within countries, while no significant correlation was observed between countries. This suggests that a person's chronotype may have a negative impact on their health outcomes only if it is later than the average chronotype of their culture. Our study contributes to resolving the inconsistent findings of previous single-country studies by indicating that the effects of chronotype on health outcomes are significant at the within-country level rather than between countries (Knutson & von Schantz, 2018; Roenneberg et al., 2019). This underscores the importance of considering an individual's deviation from their country's average chronotype as a more accurate predictor of health outcomes, emphasizing the role of cultural context in understanding the relationship between chronotype and health.

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Table 1

Number of Participants in Each Country

Countries	Sample size
Australia	254
India	255
Japan	258
Netherlands	267
South Africa	253
United States	255
Taiwan	257
South Korea	254
Colombia	260
Costa Rica	271
China	251
United Kingdom	253
Canada	260
Argentina	250
Spain	254
Belgium	255
New Zealand	252
France	253
Singapore	252
Mexico	263

Table 2

Mean and Standard Deviations of Chronotype, Physical Health, Mental Health, Depression Across Countries

	Chronotype MSFsc		**Global Physical Health *t*-score**		**Global Mental Health *t*-score**		**Depression *z*-score**	
Countries	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
Australia	937.45	113.20	54.11	7.01	41.95	8.77	0.12	1.05
India	920.90	100.37	56.88	6.79	46.95	8.22	-0.02	0.98
Japan	972.13	98.58	49.87	6.86	40.34	7.95	-0.05	1.02
Netherlands	939.27	111.47	54.20	7.27	45.48	8.80	-0.20	1.05
South Africa	896.06	84.14	54.59	6.77	42.72	8.55	0.23	1.13
United States	922.36	100.63	56.22	6.89	43.52	8.88	0	1.03
Taiwan	968.47	101.45	51.68	6.18	41.04	7.19	0.02	0.90
South Korea	962.48	107.74	51.32	6.19	41.93	7.45	-0.30	0.85
Colombia	897.98	79.88	55.19	6.43	45.44	8.11	-0.09	0.95
Costa Rica	917.2	76.73	53.93	6.69	44.91	8.81	-0.09	0.93
China	958.74	76.93	54.25	6.88	46.29	8.82	-0.32	0.78
United Kingdom	933.66	92.49	53.61	7.49	40.97	8.50	0.54	1.10
Canada	944.96	104.95	54.21	7.31	42.02	8.80	0.10	1.10
Argentina	1014.08	86.19	54.16	6.84	43.93	8.03	-0.08	0.88
Spain	992.92	121.49	53.64	6.33	43.87	7.60	-0.04	0.90
Belgium	940.07	88.45	54.44	6.63	43.40	7.86	0.04	0.97
New Zealand	908.47	82.13	54.09	7.46	43.36	8.59	0.03	1.06
France	936.19	87.14	54.47	7.40	42.08	7.85	0.23	1.02
Singapore	954.33	101.82	53.77	6.67	43.81	8.18	0.05	1.01
Mexico	970.20	89.03	54.98	6.52	45.68	7.04	-0.16	0.89
Overall	944.6	100.155	53.94	6.99	43.49	8.40	0	1

Tabel 3:

Multilevel Modeling Results for Chronotype Predicting Global Physical Health Level

Predictors	*Estimates*	*95% CI*	*p*
Fixed effects			
(Intercept)	81.14	52.25 – 110.02	**<0.001**
Chronotype (within-country)	-0.56	-0.79 – -0.33	**<0.001**
Chronotype (between-country)	-3.49	-5.60 – -1.37	0.064
gender [Man]	3.43	-4.17 – 11.04	0.376
gender [Non-binary]	-0.24	-9.85 – 9.38	0.961
age	0.04	0.01 – 0.06	**0.002**
education [Collegecompleted]	-0.44	-3.59 – 2.71	0.784
educationElementary school (K-7) completed	-1.94	-5.88 – 2.00	0.334
education [High schoolcompleted]	0.01	-3.14 – 3.16	0.994
education [Post-graduatedegree]	-0.08	-3.26 – 3.10	0.961
education [Some college]	-0.18	-3.37 – 3.02	0.914
educationSome high school (8-11)	-1.07	-4.39 – 2.25	0.526
education [Universitycourses]	1.06	-2.17 – 4.29	0.52
education [Universitydegree]	0.17	-2.95 – 3.29	0.916
Income	0.24	0.16 – 0.32	**<0.001**
Random Effects			
Residual variance	44.72		
τ_{00} country	3.9		
τ_{11} country.chronotype_dev	0		
ρ_{01} country	0.98		
ICC	0.08		
N country	20		
Observations	3848		
Marginal R^2 / Conditional R^2	0.037 / 0.115		

Note: Bolded *p*-values refer to effects that are significant at $p < 0.05$.

Table 4:

Multilevel Modeling Results for Chronotype Predicting Global Mental Health Level

Predictors	*Estimates*	*CI*	*p*
(Intercept)	56.05	22.65 – 89.45	**0.001**
Chronotype (within-country)	-0.75	-1.08 – -0.420	**<0.001**
Chronotype (between-country)	-2.19	-5.00 – 0.62	0.234
gender [Man]	4.56	-6.57 – 15.69	0.422
gender [Non-binary]	1.34	-11.52 – 14.20	0.838
age	0.04	0.02 – 0.07	**0.001**
education [Collegecompleted]	-1.18	-4.69 – 2.32	0.509
Education [Elementary school (K-7) completed	-2.20	-6.62 – 2.22	0.33
education [High schoolcompleted]	-1.30	-4.80 – 2.20	0.466
education [Post-graduatedegree]	-0.22	-3.75 – 3.31	0.904
education [Some college]	-1.73	-5.28 – 1.82	0.34
Education [Some high school (8-11)]	-4.13	-7.83 – -0.43	0.029
education [Universitycourses]	-1.01	-4.60 – 2.58	0.58
education [Universitydegree]	-0.36	-3.83 – 3.11	0.839
income	0.40	0.32 – 0.49	**<0.001**
Random Effects			
$\sigma 2$	63.99		
τ_{00} country	5.71		
τ_{11} country.chronotype_dev	0.00		
ρ_{01} country	-0.64		
ICC	0.08		
N country	20.00		
Observations	4334.00		
Marginal R^2 / Conditional R^2	0.054 / 0.134		

Note: ICC=intra-class correlation. Bolded *p*-values refer to effects that are significant at $p < 0.05$.

Tabel 5:

Multilevel Modeling Results for Chronotype Predicting Depression Level

Predictors	*Estimates*	*CI*	*p*
(Intercept)	1.18	-2.55 – 4.90	0.537
Chronotype (within-country)	0.07	0.04 – 0.11	**0.001**
Chronotype (between-country)	0.02	-0.29 – 0.25	0.715
gender [Man]	-0.17	-1.25 – 0.91	0.755
gender [Non-binary]	0.04	-1.28 – 1.37	0.948
age	-0.01	-0.01 – -0.01	**<0.001**
education [Collegecompleted]	0.22	-0.18 – 0.61	0.29
educationElementary school (K-7) completed	0.41	-0.10 – 0.92	0.118
education [High schoolcompleted]	0.23	-0.17 – 0.63	0.255
education [Post-graduatedegree]	0.18	-0.23 – 0.58	0.391
education [Some college]	0.23	-0.18 – 0.63	0.274
educationSome high school (8-11)	0.47	0.05 – 0.89	0.029
education [Universitycourses]	0.24	-0.17 – 0.65	0.247
education [Universitydegree]	0.14	-0.25 – 0.54	0.481
income	-0.04	-0.05 – -0.03	**<0.001**
Random Effects			
Residual Variance	0.9		
τ_{00} country	0.08		
τ_{11} country.chronotype_dev	0		
ρ_{01} country	-0.74		
ICC	0.09		
N country	20		
Observations	4333		
Marginal R^2 / Conditional R^2	0.053 / 0.135		

Note: ICC=intra-class correlation. Bolded *p*-values refer to effects that are significant at $p < 0.05$.

Figure 1

The Effect of Chronotype on Global Mental Health, Globel Physical Health and Depression Within and Between Countries

